

Statistical Analysis Report

CDC WONDER Heart Disease Mortality Disparities

United States, 2020–2024

Diseases of Heart (ICD-10: I00–I09, I11, I13, I20–I51)

By Race, Sex, and Census Region

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Version	1.0
Status	Final
Software	SAS Studio (SAS OnDemand for Academics)
Repository	github.com/alemikhailov/cdc-wonder-mortality

1. Purpose

This report documents the statistical analysis of heart disease mortality disparities in the United States from 2020 to 2024. The analysis uses data from the CDC WONDER Underlying Cause of Death database to examine differences in heart disease mortality rates across racial/ethnic groups, sex, and US Census regions.

The objectives of this analysis are to: (1) describe the distribution of heart disease mortality by demographic and geographic characteristics; (2) identify statistically significant disparities in mortality rates between racial/ethnic groups, adjusted for sex, region, and calendar year; and (3) document data sources, validation steps, statistical methods, results, limitations, and public health interpretation in a reproducible, auditable format.

2. Data Source

Item	Value
Database	CDC WONDER Underlying Cause of Death, 2018–2024, Single Race
URL	https://wonder.cdc.gov/ucd-icd10-expanded.html
ICD-10 Codes	I00–I09, I11, I13, I20–I25, I26–I28, I30–I52 (Diseases of Heart)
Years	2020–2024
Group By	Census Region, Single Race 6, Sex, Year
Export Format	Tab-separated values (.tsv)
Access Date	September 2026

The dataset contains aggregate counts of deaths, population denominators, and crude mortality rates per 100,000 population for each combination of Census region, race/ethnicity, sex, and calendar year. CDC WONDER suppresses cells with fewer than 10 deaths to protect individual privacy. The six racial/ethnic categories available in this dataset are: American Indian or Alaska Native, Asian, Black or African American, More than one race, Native Hawaiian or Other Pacific Islander, and White.

3. Data Processing

3.1 Import (01_import.sas)

The raw CDC WONDER export was imported into SAS using a DATA step with tab delimiter specification (hex 09), DSD option for quoted text fields, and MISSOVER to handle varying record lengths. All numeric fields were read as character and converted to numeric using the INPUT function with the ?? modifier to suppress conversion notes for missing values in subtotal rows.

Subtotal and total rows inserted by CDC WONDER (identified by the Notes field containing "Total" or missing Year values) were excluded. A log-population offset variable was derived for Poisson regression modeling. All variables were assigned descriptive labels and appropriate formats. The raw import contained 390 records; after exclusions, the analysis dataset contained 238 records.

3.2 Validation (02_validate.sas)

Six validation checks were performed on the cleaned dataset before analysis:

Check	Result	Detail
Record count	PASS	238 of 240 expected records present (6 races × 2 sexes × 4 regions × 5 years = 240)
Duplicate records	PASS	No duplicate Region × Race × Sex × Year combinations found
Missing values	PASS	No missing values in key variables (Deaths, Population, Crude Rate, Year)
Value range checks	PASS	No negative death counts, no invalid populations, all years within 2020–2024
Confidence interval consistency	PASS	Rate_LCL ≤ Crude_Rate ≤ Rate_UCL for all records
Rate recalculation	PASS	All CDC-provided rates match (Deaths / Population) × 100,000 within ±0.5

3.3 Suppressed Data

Two records were suppressed by CDC WONDER due to small cell counts (fewer than 10 deaths): Native Hawaiian or Other Pacific Islander females in the Northeast region for two of the five study years. This suppression accounts for the shortfall from the expected 240 records to the observed 238. The affected stratum (NHOPI females, Northeast) retains three years of data and is included in the analysis with the understanding that estimates for this group in this region carry wider uncertainty.

4. Statistical Methods

4.1 Descriptive Analysis

Descriptive tables were produced using PROC TABULATE showing mean crude mortality rates and total death counts by race and sex (Table 1), by region and race (Table 2), and by race and year (Table 3). Visualizations were generated using PROC SGPLOT and PROC SGPPANEL including grouped bar charts, trend lines, and regional comparison panels.

4.2 Poisson Regression (Model 1)

A Poisson regression model was fit using PROC GENMOD to estimate adjusted mortality rate ratios by race, controlling for sex, Census region, and calendar year. The model specified Deaths as the dependent variable with a log link function and log(Population) as an offset, which models rates (deaths per person) rather than raw counts. White was selected as the reference category for race, and Female as the reference for sex. ESTIMATE statements with the EXP option were used to produce exponentiated rate ratios with 95% confidence intervals.

4.3 Interaction Model (Model 2)

A second Poisson regression model included a Race × Sex interaction term to test whether racial disparities in heart disease mortality differ between men and women. The Type 3 likelihood ratio test for the interaction term assessed the overall significance of effect modification by sex.

5. Results

5.1 Descriptive Findings

The analysis dataset included 238 records representing 3,531,898 heart disease deaths across 5 years (2020–2024), 4 Census regions, 6 racial/ethnic groups, and 2 sexes. White populations accounted for the largest share of deaths, consistent with their larger population denominators. Crude mortality rates were highest among White and Black or African American populations, while More than one race and Asian populations had the lowest crude rates. Males had consistently higher crude mortality rates than females across all racial/ethnic groups and regions.

5.2 Poisson Regression — Rate Ratios (Model 1)

All race and sex effects were statistically significant ($p < 0.0001$). Adjusted rate ratios relative to White were:

Comparison	Rate Ratio	95% LCL	95% UCL	P-value
AIAN vs White	0.40	0.394	0.405	<0.0001
Asian vs White	0.38	0.381	0.386	<0.0001
Black or African American vs White	0.86	0.853	0.858	<0.0001
More than one race vs White	0.15	0.147	0.152	<0.0001
NHOPI vs White	0.57	0.552	0.583	<0.0001
Male vs Female	1.25	1.250	1.255	<0.0001

All non-White racial/ethnic groups had significantly lower adjusted heart disease mortality rates than White populations. Black or African American populations had the smallest disparity (rate ratio 0.86, 14% lower), while More than one race had the largest (rate ratio 0.15, 85% lower). Males had 25% higher mortality than females (rate ratio 1.25). Heart disease mortality showed a small but statistically significant decline over the study period (Year coefficient: -0.012 , $p < 0.0001$).

5.3 Model Fit

The Deviance/DF statistic was 105.6, substantially above 1.0, indicating significant overdispersion. This is expected for aggregate population-level mortality data where death counts within demographic strata exhibit more variability than the Poisson distribution assumes. A negative binomial model or a Poisson model with a scale parameter could address this overdispersion; however, the primary purpose of this analysis is to estimate adjusted rate ratios rather than to perform hypothesis tests at nominal significance levels, and the point estimates remain valid under overdispersion.

6. Limitations

Crude rates without age adjustment. The mortality rates provided by CDC WONDER in this query are crude rates (deaths per 100,000 total population), not age-adjusted rates. Because heart disease disproportionately affects older adults, and racial/ethnic groups differ substantially in age distribution, crude rates primarily reflect the age structure of each population rather than underlying disease risk. The finding that White populations have the highest crude rates is largely attributable to the older age distribution of the White population. Age-adjusted rates would likely show a different pattern, particularly for Black or African American populations, which have well-documented elevated cardiovascular mortality when age is accounted for. This limitation is the single most important caveat for interpreting these results.

Suppressed cells. Two records were suppressed by CDC WONDER for small cell sizes. This affects the Native Hawaiian or Other Pacific Islander female population in the Northeast and may introduce slight bias in estimates for this group.

Overdispersion. The Poisson model exhibits significant overdispersion (Deviance/DF = 105.6). While point estimates of rate ratios are unaffected, standard errors may be underestimated, which means confidence intervals may be too narrow and p-values too small. For a production analysis, a negative binomial model or scaled Poisson model would be appropriate.

Single Race classification. The CDC WONDER Single Race 6 classification does not include Hispanic/Latino as a separate category, as Hispanic origin is considered an ethnicity rather than a race. Individuals of Hispanic origin are distributed across the six racial categories. This limits the ability to examine Hispanic/Latino mortality disparities, which are an important public health concern.

Ecological analysis. This is an aggregate-level analysis based on population-level mortality counts and denominators. Individual-level confounders such as socioeconomic status, smoking, comorbidities, and access to healthcare cannot be controlled for. Associations observed at the group level may not hold at the individual level (ecological fallacy).

7. Programs and Reproducibility

Program	Purpose
01_import.sas	Import raw TSV export, convert variable types, exclude subtotals, create analysis dataset
02_validate.sas	Record count, duplicate, missing value, range, CI consistency, and rate recalculation checks
03_analysis.sas	Descriptive tables (PROC TABULATE), visualizations (PROC SGPLOT/SGPANEL), Poisson regression (PROC GENMOD)

All programs are maintained under version control in a public GitHub repository. The raw data file, all SAS programs, and this report are available for review and reproduction. Programs should be executed in numeric order (01, 02, 03) within a single SAS session, as subsequent programs depend on datasets created by earlier steps.

8. Environment

Component	Version
SAS	SAS OnDemand for Academics (SAS Studio)
Operating System	Server-hosted (Linux)
Version Control	Git / GitHub